bketLock 2.0 SDK Usage

The bketLock SDK provides one jar package and two jni .so files (32-bit and 64-bit):

```
bket_lock_2.0.0.jar
lib----liblock.so(armeabi-v7a)    32-bit machine
lib----libbklock.so(armeabi-v7a)  32-bit machine
lib----liblock.so(arm64-v8)       64-bit machine
lib----libbklock.so(armeabi-v8)   64-bit machine
```

Change Log

#	Revision Content	Version	Release Date	Author
1	Released document	V2.0.0	2024-12-14	Qiu Peng
2	1. Added electromagnetic lock power control interface	V2.0.2	2025-02-21	Qiu Peng
3	1. Fixed lock information reporting	V2.0.4	2025-03-03	Qiu Peng
4	1. Modified model compatibility	V2.0.6	2025-03-11	Qiu Peng
5	1. Added set-lock-door-detection interface 2. Added "No SIM card" option to network status setting	V2.0.8	2025-03-20	Qiu Peng

I. Usage Instructions

1. SDK Library Files

The bketLock SDK provides one jar package and two jni .so files (32-bit and 64-bit):

```
bket_lock_2.0.0.jar
lib----liblock.so(armeabi-v7a)    32-bit machine
lib----libbklock.so(armeabi-v7a)  32-bit machine
lib----liblock.so(arm64-v8)       64-bit machine
lib----libbklock.so(armeabi-v8)   64-bit machine
```

2. Importing the Library into the Project

- Place the jar in the `libs` directory of the integration project Module, reference the imported jar in the Module's `build.gradle`, then perform a Gradle sync of the whole project.

```
dependencies {
    implementation files('libs\\bket_lock_2.0.0.jar')
}
```

- Copy the .so files into the project's jniLibs:

```
app\src\main\jniLibs
```

3. SDK Initialization

```
import com.bket.bketlock.BketLockAdapter;

public class MainActivity extends Activity {
    private BketLockAdapter mBketLockAdaper;

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        mBketLockAdaper = BketLockAdapter.getInstance(getApplicationContex
        String version = mBketLockAdaper.readSdkVersion();
        Log.d(TAG, "bkLock version " + version);
    }
}
```

4. Releasing the SDK

```
protected void onDestroy(){
    super.onDestroy();
    mBketLockAdaper.release();
    Log.d(TAG, "onDestroy: quit");
}
```

5. Unlocking Methods

5.1 Individual Lock/Unlock

> Individual lock/unlock interfaces: openLock1 / closeLock1 / openLock2 / closeLock2
>
> When using the individual lock/unlock interfaces, you need to poll `getDoorAndLockStatus` to query the current state fed back by the door and lock, and handle the logic accordingly.

5.2 Automatic Lock/Unlock

> For the automatic lock/unlock flow, call the `autoLock1Open` interface, and handle it based on the status in the `autoOpenLockStatusIface` callback.
>
> `setLockOpenWaitTime` sets the door-open wait timeout and door-close wait timeout for this lock/unlock flow.

6. Notes

II. Interface Description

1. Lock Control

1.1 Open Lock 1

```
openLock1
public void openLock1(void)
```

Open lock 1

Parameters: none

Returns: none

1.2 Close Lock 1

```
closeLock1
public void closeLock1(void)
```

Close lock 1

Parameters: none

Returns: none

1.3 Open Lock 2

```
openLock2
public void openLock2(void)
```

Open lock 2

Parameters: none

Returns: none

1.4 Close Lock 2

```
closeLock2
public void closeLock2(void)
```

Close lock 2

Parameters: none

Returns: none

1.5 Lock Power On

```
setDevDcsPowerOn
public void setDevDcsPowerOn(int Index)
```

Electromagnetic lock power on

Parameters: Index - no meaning, default 0

Returns: none

1.6 Lock Power Off

```
setDevDcsPowerOff
public void setDevDcsPowerOff(int deviceIndex)
```

Electromagnetic lock power off

Parameters: Index - no meaning, default 0

Returns: none

1.7 Lock Status

1.7.1 Read Lock Status

```
getDoorAndLockStatus
public java.util.List<java.lang.Integer> getDoorAndLockStatus(int index)
```

Parameters: index - currently no meaning, default 0

Returns: Integer List

Index	Meaning
0	Lock 1 status
1	Door 1 status

2	Whether lock 1 is online
3	Lock 2 status
4	Door 2 status
5	Whether lock 2 is online

1.7.2 Lock Status Callback

Callback interface

```
lockDoorStatus
void lockDoorStatus(int lock, int door, int lockOnline, long timeSp)
```

Parameters:

```
lock        0 lock closed   1 lock open
door        0 door closed   1 door open
lockOnLine  0 lock offline  1 lock online
timeSp      timestamp
```

Add lock 1 callback

```
addLock1StatusListenerCallback
public void addLock1StatusListenerCallback(LockDoorStatusIface callback)
```

Add lock control status callback. Parameters: callback - LockDoorStatusIface callback class

Remove lock 1 callback

```
removeLock1StatusListenerCallback
public void removeLockStatusListenerCallback(LockDoorStatusIface callback)
```

Remove lock control status callback. Parameters: callback - LockDoorStatusIface callback class

Add lock 2 callback

```
addLock2StatusListenerCallback
public void addLock1StatusListenerCallback(LockDoorStatusIface callback)
```

Add lock control status callback. Parameters: callback - LockDoorStatusIface callback class

Remove lock 2 callback

```
removeLock2StatusListenerCallback
public void removeLockStatusListenerCallback(LockDoorStatusIface callback)
```

Remove lock control status callback. Parameters: callback - LockDoorStatusIface callback class

1.8 Lock Information --- Extended Information Interface

Callback interface

```
lockInfo
void lockInfo(int state, int hall, int hallNullVal, int hallThreVal, long t
```

Parameters:

```
state        status flag bit
hall         real-time Hall value
hallNullVal  Hall no-load value
hallThreVal  Hall threshold value
timeSp       timestamp
```

Add lock 1 info callback

```
addLock1InfoListenerCallback
public void addLock1InfoListenerCallback(LockInfoCallback callback)
```

Parameters: callback - LockInfoCallback callback class

Add lock 2 info callback

```
addLock2InfoListenerCallback
public void addLock2InfoListenerCallback(LockInfoCallback callback)
```

Parameters: callback - LockInfoCallback callback class

Remove lock 1 info callback

```
removeLock1InfoListenerCallback
public void removeLock1InfoListenerCallback(LockInfoCallback callback)
```

Parameters: callback - LockInfoCallback callback class

Remove lock 2 info callback

```
removeLock2InfoListenerCallback
public void removeLock2InfoListenerCallback(LockInfoCallback callback)
```

Parameters: callback - LockInfoCallback callback class

1.9 Automatic Lock/Unlock Control

1.9.1 Set Automatic Lock/Unlock Flow Delay

```
setLockOpenWaitTime
public int setLockOpenWaitTime(int openWaitTime, int CloseWaitTime);
```

Parameters:

```
int openWaitTime   Max wait delay to open the door after unlocking. Defaul
int CloseWaitTime  Max wait delay to close door/lock. Default 360s, range >
                   and CloseWaitTime must be more than 15s greater than ope
                   (the threshold boundaries are exclusive).
```

Correct example: openWaitTime = 15, CloseWaitTime = 300

Incorrect examples:
openWaitTime = 67, CloseWaitTime = 300 (openWaitTime out of range)
openWaitTime = 30, CloseWaitTime = 40 (CloseWaitTime - openWaitTime < 15)

Returns:

```
 0  Set successfully
-1  Door-open wait time does not meet requirements
-2  Door-close wait time does not meet requirements
```

1.9.2 Automatic Lock/Unlock Callback

Callback interface

```
autoOpenLockStatusIface
void autoOpenLockStatus(String status, long timeSp)
```

Parameters: status; timeSp - timestamp. For common flows see "3.3 Common Callback States".

status	Description
0	Lock and door states are both normal; unlocking is allowed
1	Normal door/lock close; open-close flow complete; current flow ends
2	Lock and door states abnormal; no unlock; current flow ends
3	After unlocking, lock state detected abnormal; flow ends
4	Auto lock/unlock flow exceeded the set maximum timeout (default 6 min); flow ends
5	Previous flow not finished and called again; current flow ends
6	Lock opened normally, but door not opened
7	Lock opened normally and door opened normally
8	Received the calling instruction confQueryElectromagneticLock1Status
9	After unlocking, lock feedback "not open" but door is open; marked as abnormal door-open transaction event
A	After door-open timeout, door closed and no door-open detected

1.9.3 Lock 1 Automatic Lock/Unlock

```
autoLock1Open
public void autoLock1Open(autoOpenLockStatusIface callback);
```

Parameters: callback - autoOpenLockStatusIface callback class. Returns: none

1.9.4 Lock 2 Automatic Lock/Unlock

```
autoLock2Open
public void autoLock2Open(autoOpenLockStatusIface callback);
```

Parameters: callback - autoOpenLockStatusIface callback class. Returns: none

1.10 Set Whether Lock 1 Has Door Detection

```
setLock1Type
public void setLock1Type(int DoorType)
```

Parameters: DoorType = 1 detect, = 2 do not detect. Returns: none

1.11 Set Whether Lock 2 Has Door Detection

```
setLock2Type
public void setLock2Type(int DoorType)
```

Parameters: DoorType = 1 detect, = 2 do not detect. Returns: none

1.12 Read Whether Lock Has Door Detection

```
getLockDoorType
public java.util.List<java.lang.Integer> getLockDoorType(int index)
```

Parameters: index - currently no meaning, default 0. Returns: Integer List

```
0  Lock 1 door detection state --- 0 detect, 1 do not detect
1  Lock 2 door detection state --- 0 detect, 1 do not detect
```

2. Power Control

2.1 Power Supply Status

2.1.1 Read Power Supply Status

```
getPowerStatus
public int getPowerStatus();
```

Parameters: none. Returns: power supply status

```
d1 power status = 0 battery, = 1 mains, = 2 unknown
```

2.1.2 Power Supply Status Listening

Callback interface

```
powerStatus
void powerStatus(int status, long timeSp)
```

Parameters: status - power status, 0 battery supply, 1 power supply; timeSp - timestamp

Add callback

```
addPowerStatusListenerCallback
public void addPowerStatusListenerCallback(PowerStatusIface callback)
```

Add power status change callback. Parameters: callback - PowerStatusIface callback class

Remove callback

```
removePowerStatusListenerCallback
public void removePowerStatusListenerCallback(PowerStatusIface callback)
```

Remove power status change callback. Parameters: callback - PowerStatusIface callback class

2.2 Set Power-Down Delay Shutdown Time

```
setPowerDownTime
public void confPowerDownTime(int time);
```

Parameters: int time - power-down delay working time, in minutes. Returns: none

Description: Sets the battery-powered working time after mains power is cut. Default 10 minutes.

2.3 Shutdown

```
setPowerOff
public void setPowerOff();
```

X18 X16 mains-supply reboot, battery-supply power off

X10 mains supply invalid, battery-supply power off

Parameters: none. Returns: none

2.4 Reboot

```
setPowerReboot
public void setPowerReboot();
```

X18 X16 mains-supply reboot, battery-supply power off

X10 mains supply invalid, battery-supply power off

Parameters: none. Returns: none

2.5 12V Output Control

2.5.1 12V Output Power Control and Status Acquisition

```
setOutputPower
public void setOutputPower(int data);
```

Parameters: int data - 0 turn off power output, 1 turn on power output. Returns: none

3. Common Functions

3.1 Get Class Handle

```
BketLockAdapter getInstance(Context context);
```

Parameters: context - Activity or Service context handle. Returns: none

3.2 Version Number

3.2.1 Get SDK Version Number

```
getSdkVersion
public String getSdkVersion()
```

Parameters: none. Returns: String SDK version number "BketLock_SDK_V2.0.0_20241214"

3.2.2 Get Control Board Version Number

```
getLockDriverVersion
public String getLockDriverVersion();
```

Parameters: none. Returns: String version number "BKX10_V1.0.0_20240821"

3.3 Watchdog

3.3.1 Set Watchdog

```
setWatchDog
public void setWatchDog(int type, portCallback callback);
```

Parameters: int type - 1 enable watchdog, 0 disable watchdog; portCallback callback - set-result callback. Returns: none

3.4 Network Status

3.4.1 Set Network Status

```
setNetStatus
public void setNetStatus(int data);
```

Parameters:

```
int data  01  no network
          02  has network
          03  no SIM card
```

Returns: none

3.5 Control Board Upgrade

3.5.1 Upgrade Interface

```
confUpdateFirmware
public int confUpdateFirmware(String firmwareFilePath);
```

Parameters: String firmwareFilePath - full path of the upgrade package

Returns: int 0 success, non-zero failure

```
private final int UPDATE_FAIL_NO_FILE = -1;
private final int UPDATE_FAIL_READ_VERSION = -2;
private final int UPDATE_FAIL_CHECK_VERSION = -3;
private final int UPDATE_FAIL_JUMP_TO_BOOT = -4;
private final int UPDATE_FAIL_READ_BOOT_VERSION = -5;
private final int UPDATE_FAIL_IAP_ERASE = -6;
private final int UPDATE_FAIL_IAP_ROM = -7;
private final int UPDATE_FAIL_JUMP_TO_APP = -8;
private final int UPDATE_FAIL_UNZIP_UPDATE_FILE = -9;
private final int UPDATE_FAIL_SAME_VERAION = -10;
```

> **Upgrade notes**
>
> 1. After the control board upgrade succeeds, the system will reboot. A normal upgrade takes 50~70s, depending on the upgrade file size.

> 2. The control board upgrade must be done during idle time, when there is no active business order.

4. Callback Interfaces

4.1 Common Interface Callback

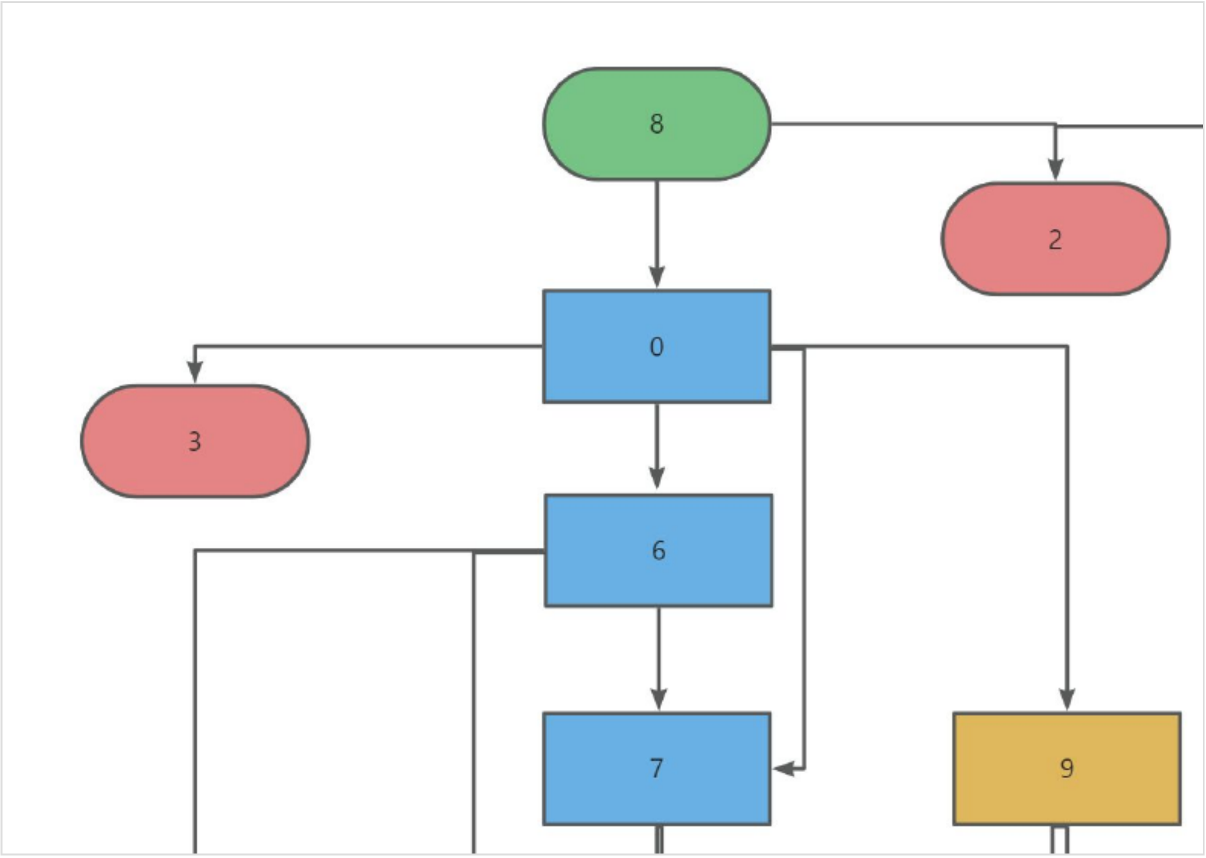
```
BketLockAdapter.portCallback

onSuccess
void onSuccess(java.lang.String result)
  Parameters: result - reply message

onFailure
void onFailure(int errorCode, java.lang.String errorMsg)
  Parameters: errorCode - error code; errorMsg - error message
```

III. Detailed Description of the Automatic Lock/Unlock Flow

3.1 Lock/Unlock Flow Call Description

Call `autoLock1Open` / `autoLock2Open` to start one automatic lock/unlock flow. If re-entered before the flow ends, it will return 8 and 5 once.

3.2 Callback Status Flow Diagram



3.3 Common Callback States

3.3.1 Normal unlock / door-open / door-close / lock flow

```
8---0---6---6......---7---1
```

3.3.2 Door opened immediately after unlocking

```
8---0---7---1
```

3.3.3 Normal unlock, door not opened, then lock flow

```
8---0---6---6...---6---1
```

3.3.4 Normal unlock, door open/close timeout

```
8---0---6---6...---7---4
```

3.3.5 Lock did not open during open/close/lock flow (abnormal lock feedback)

```
8---0---9---1
```

3.3.6 Repeated call before flow ended

```
8---5
```

3.3.7 Door or lock abnormal, cannot unlock

```
8---2
```

3.3.8 Door/lock state normal but cannot unlock

```
8---0---3
```

3.3.9 Abnormal door feedback; after door-open timeout, door closed and no door-open detected

```
8---0---6---6...6---A
```

3.3.10 After unlocking, door not opened and door-close/lock timeout

```
8---0---6---6...6---4
```